

Closed-Loop BCI-Triggered System with Soft-Robotic Glove for after-stroke rehabilitation

Flavio Massaroni

1 Clinical Need

Stroke represents a major cause of long-term neurological disability globally. As evidenced by epidemiological data from the World Stroke Organization [1], the socio-health impact of this pathology is constantly growing: over 93 million cases prevalent, some 160 million years of life lost to disability (DALYs) and a cost estimated global economic of more than US\$ 890 billion.

Between 55% and 75% of survivors have chronic upper limb motor deficit which compromises autonomy in daily activities (ADL) [12]. Donkor [2] contextualize this figure in the perspective of quality of life: stroke is the second cause of death at the world and causes severe permanent disability in about half of the survivors, with high impact on autonomy, working life and the care burden on family members.

This combination of high prevalence and high residual disability justifies the search for advanced technological solutions that can deliver portable therapy with an intensity comparable to that of departments of specialized rehabilitation. Recent clinical research indicates that the effectiveness of traditional rehabilitation treatments is severely limited by a therapeutic dosage deficit. The systematic analysis conducted by Newton et al. [3] highlights how the actual time spent on active therapeutic exercise (time on task) within conventional sessions varies on average between 8 and 44 minutes per day — significantly lower than recommended doses to induce cortical reorganization stable. A clinical paradox emerges: patients with more severe impairment receive doses of significantly lower active therapy precisely because of the high effort required without tools modular assistance.

2 Current Standard of Care and Its Limitations

The current standard of post-stroke upper limb rehabilitation combines conventional physiotherapy with commercially available robotic orthoses. Both approaches share a critical failure mode: they address the peripheral component of motor recovery while neglecting its central requirement.

Newton et al. [3] quantify this dose gap: effective active therapy time in conventional care averages 8–44 minutes per day — far below the repetition volume required to induce

stable cortical reorganization. After discharge, most patients receive virtually no structured rehabilitation.

The robotic hardware available today does not close this gap. Wang et al. [7] compared a passive soft-robotic glove against repetitive transcranial magnetic stimulation (rTMS) in patients with severe upper limb dysfunction. The passive glove improved Fugl-Meyer Assessment scores (FMA-UE) by facilitating joint mobilization, but produced no significant changes in Resting Motor Threshold (RMT) or Motor Evoked Potential (MEP) amplitude — the markers of cortical synaptic excitability. The glove moves the hand; it does not engage the brain.

The neurobiological explanation is well established. Long-term motor recovery depends on Hebbian plasticity (Spike-Timing-Dependent Plasticity, STDP): stable synaptic modifications require temporal coincidence — within approximately 300 ms — between the pre-synaptic signal (motor cortex activation associated with movement intention) and the post-synaptic afferent feedback generated by the actual limb movement [4, 5]. In the μ (8–12 Hz) and β (13–30 Hz) bands, Motor Imagery manifests as event-related desynchronization (ERD) over contralateral motor areas (C3/C4). When a BCI detects this ERD and uses it as an instantaneous trigger, the Hebbian coincidence window is realized. A random or sham activation produces no significant increase in β -band corticomuscular coherence [5] and no relevant clinical improvement [6].

The biomarker of this reorganization is measurable in real time: Rustamov et al. [11] demonstrate that theta-gamma cross-frequency coupling (CFC) over C3/C4 increases proportionally to motor recovery across BCI sessions, confirming CFC as a neurobiological proxy for treatment efficacy. The mechanism exists, is validated, and is quantifiable. The gap is not biological — it is a delivery problem.

3 The Human Limitation Being Addressed

The mechanism described above requires three conditions that no human operator can reliably provide in an unsupervised home setting:

(1) Continuous presence. The Hebbian contingency must be enforced at every repetition across sessions of 30–45 minutes, 3–5 times per week, for months. A trained therapist cannot be physically present at the patient’s home at this frequency — this is precisely the dose gap documented by Newton et al. [3]: 8–44 minutes of active therapy per day in conventional care, with near-zero structured rehabilitation after discharge.

(2) Sub-300 ms reactive latency. The BCI trigger must fire within the Hebbian coincidence window [5]. No human observer can reliably detect ERD onset in a raw EEG signal and manually trigger a mechanical response within this window, consistently, across hundreds of repetitions per session.

(3) Operator-independent consistency. Outcome variability in conventional rehabilitation is partially attributable to inter-therapist inconsistency in task delivery and feedback

timing [3]. A closed-loop BCI system applies identical contingency criteria at every trial, eliminating this source of variance. Ren et al. [9] confirm in a meta-analysis of 10 RCTs (290 patients) that BCI-triggered intervention outperforms conventional therapy (SMD = 0.50, 95 % CI: 0.26–0.73), with effect size directly linked to contingency fidelity.

The robot is not a substitute for clinical judgment — the therapist remains in the loop asynchronously via remote dashboard. It is a substitute for physical presence and millisecond-level reactivity: engineering problems, not clinical ones.

Referenser

- [1] Valery L. Feigin, Michael Brainin, Bo Norrving, Sheila O. Martins, Jeyaraj Pandian, Patrice Lindsay, Maria F. Grupper, and Ilari Rautalin. World stroke organization: Global stroke fact sheet 2025. *International Journal of Stroke*, 20(2):132–144, 2025.
- [2] Emmanuel S. Donkor. Stroke in the 21st century: A snapshot of the burden, epidemiology, and quality of life. *Stroke Research and Treatment*, 2018:3238165, 2018.
- [3] Sarah P. Newton, Emily J. Dalton, Jia Y. Ang, Marlena Klaic, Vincent Thijs, and Kathryn S. Hayward. Dose, content, and context of usual care in stroke upper limb motor interventions: A systematic review. *Clinical Rehabilitation*, 37(11):1437–1450, 2023.
- [4] A. Biasiucci, R. Leeb, I. Iturrate, S. Perdikis, A. Al-Khodairy, T. Corbet, A. Schnider, T. Schmidlin, H. Zhang, M. Bassolino, D. Viceic, P. Vuadens, A. G. Guggisberg, and J. d. R. Millán. Brain-actuated functional electrical stimulation elicits lasting arm motor recovery after stroke. *Nature Communications*, 9:2421, 2018.
- [5] Johanna Krueger, Richard Krauth, Christoph Reichert, Serafeim Perdikis, Susanne Vogt, Tessa Huchtemann, Stefan Dürschmid, Almut Sickert, Juliane Lamprecht, Almir Huremovic, Michael Görtler, Slawomir J. Nasuto, I.-Chin Tsai, Robert T. Knight, Hermann Hinrichs, Hans-Jochen Heinze, Sabine Lindquist, Michael Sailer, Jose del R. Millán, and Catherine M. Sweeney-Reed. Hebbian plasticity induced by temporally coincident bci enhances post-stroke motor recovery. *Scientific Reports*, 14:18700, 2024.
- [6] Myeong Sun Kim, Hyunju Park, Ilho Kwon, Kwang-Ok An, Hayeon Kim, Gyulee Park, Wooseok Hyung, Chang-Hwan Im, and Joon-Ho Shin. Efficacy of brain-computer interface training with motor imagery-contingent feedback in improving upper limb function and neuroplasticity among persons with chronic stroke: a double-blinded, parallel-group, randomized controlled trial. *Journal of NeuroEngineering and Rehabilitation*, 22:1, 2025.
- [7] Taotao Wang, Zhonghua Liu, Jianxiong Gu, Jizhi Tan, and Tian Hu. Effectiveness of soft robotic glove versus repetitive transcranial magnetic stimulation in post-stroke patients with severe upper limb dysfunction: A randomised controlled trial. *Frontiers in Neurology*, 13:887205, 2023.
- [8] Bethel A. C. Osuagwu, Sarah Timms, Ruth Peachment, Sarah Dowie, Helen Thrussell, Susan Cross, Rebecca Shirley, Antonio Segura-Fragoso, and Julian Taylor. Home-based rehabilitation using a soft robotic hand glove device leads to improvement in hand function in people with chronic spinal cord injury: a pilot study. *Journal of NeuroEngineering and Rehabilitation*, 17:40, 2020.
- [9] Chunlin Ren, Xinmin Li, Qian Gao, Mengyang Pan, Jing Wang, Fangjie Yang, Zhenfei Duan, Pengxue Guo, and Yasu Zhang. The effect of brain-computer interface controlled

functional electrical stimulation training on rehabilitation of upper limb after stroke: a systematic review and meta-analysis. *Frontiers in Human Neuroscience*, 18:1438095, 2024.

- [10] Zhen-Zhen Ma, Jia-Jia Wu, Zhi Cao, Xu-Yun Hua, Mou-Xiong Zheng, Xiang-Xin Xing, Jie Ma, and Jian-Guang Xu. Motor imagery-based brain-computer interface rehabilitation programs enhance upper extremity performance and cortical activation in stroke patients. *Journal of NeuroEngineering and Rehabilitation*, 21:91, 2024.
- [11] Nabi Rustamov, Lauren Souders, Lauren Sheehan, Alexandre Carter, and Eric C. Leuthardt. Ipsihand brain-computer interface therapy induces broad upper extremity motor rehabilitation in chronic stroke. *Neurorehabilitation and Neural Repair*, 39(1):74–86, 2025.
- [12] Tommaso Proietti, Kristin Nuckols, Jesse Grupper, Diogo Schwerz de Lucena, Bianca Inirio, Kelley Porazinski, Diana Wagner, Tazzy Cole, Christina Glover, Sarah Mendelowitz, Maxwell Herman, Joan Breen, David Lin, and Conor Walsh. Combining soft robotics and telerehabilitation for improving motor function after stroke. *Wearable Technologies*, 5:e1, 2024.
- [13] Shizhe Wu, Kinkini Bhadra, Anne-Lise Giraud, and Silvia Marchesotti. Adaptive lda classifier enhances real-time control of an eeg brain-computer interface for decoding imagined syllables. *Brain Sciences*, 14:196, 2024.
- [14] Brian Premchand, Zhuo Zhang, Kai Keng Ang, Juanhong Yu, Isaac Okumura Tan, Josephine Pei Wen Lam, Anna Xin Yi Choo, Ananda Sidarta, Patrick Wai Hang Kwong, and Lau Ha Chloe Chung. A personalized multimodal bci-soft robotics system for rehabilitating upper limb function in chronic stroke patients. *Biomimetics*, 10:94, 2025.
- [15] Vasiliki Fiska, Konstantinos Mitsopoulos, Vasiliki Mantiou, Vasileia Petronikolou, Panagiotis Antoniou, Konstantinos Tagaras, Konstantinos Kasimis, Konstantinos Nizamis, Markos G. Tsipouras, Alexander Astaras, Panagiotis D. Bamidis, and Alkinoos Athanasiou. Integration and validation of soft wearable robotic gloves for sensorimotor rehabilitation of human hand function. *Applied Sciences*, 15:5299, 2025.
- [16] Yiqing Lu, Weiwei Yang, Song Wu, Yicheng Li, Jinhu Wei, Ming Li, Yongcheng Li, and Yaping Huai. Exploring neural activity changes during motor imagery-based brain-computer interface training with robotic hand for upper limb rehabilitation in ischemic stroke patients: a pilot study. *Frontiers in Human Neuroscience*, 19:1626000, 2025.